

Multi – Modal Cyberbullying Detection using Deep Learning and NLP Techniques

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Abstract—Cyberbullying is one of the most challenging problems in today's digital world. With the proliferation of various social media, video-sharing platforms, and online forums, psychological and social detriments to victims caused by cyberbullying have scaled new heights. Methods traditionally developed for detecting cyberbullying can only handle single modalities, and fail to capture the complex and subtle manifestations of cyberbullying across text, image, and video, calling for an urgent, more comprehensive solution. This paper proposes a multi-modal cyberbullying detection framework to analyze user-generated content with advanced deep learning methods and natural language processing from three modalities. For video analysis, it uses YOLOv8 to detect bullying behaviors and injurious interactions from frames in real-time; classify aggressive visual contents by fine-tuned MobileNetV2; use NLP models to detect abusive language and threats from text. Further, a fusion mechanism integrates these outputs to improve detection accuracy, robustness, and scalability. This is the first unified, real-time, multi-modal system to provide a holistic and effective solution for a safer online environment.

Keywords—Online Safety, Social Media Moderation, Abusive Content, Multimedia Analysis, User Behaviour, Digital Harassment.

I. INTRODUCTION

Digital platforms have changed how people interact, affording unparalleled opportunities for social connectivity, information dissemination, and creative expression. On the flip side, however, they have also created a host of social ills, with cyberbullying being one of the most alarming issues to escalate in online environments today. Quite unlike traditional bullying, this is a type of harassment [1] that occurs across electronic media, often without revealing the identity of the perpetrator, and can target victims anywhere, at any time. With many dire consequences, including psychological distress, social isolation, and in extreme cases, long-term mental illness, cyberbullying poses grave concerns.

The widespread nature of online platforms, including social media networks, video-sharing sites, and messaging apps, has escalated the rate of potentially damaging interactions, making reliable detection mechanisms an urgent need to safeguard users, particularly adolescents who are susceptible to such threats. Cyberbullying is inherently complex and multi-dimensional; it includes not only offensive language [2] but also threatening messages, coordinated harassment, and aggressive visual content. This variability in expression greatly complicates the task of automatic detection, since these harmful behaviors can manifest in multiple forms of media, including textual comments, images, and video sequences.

Traditional unimodal approaches to detection, such as analyzing either text or images, fail to capture the context surrounding an abusive behavior and result in limited accuracy and a higher number of false negatives. Furthermore, subtle forms of bullying, such as sarcasm, coded language, and display of aggressive gestures in a video, are often missed by a unimodal approach. Such a limitation mandates a holistic approach that jointly analyzes various modalities through which cyberbullying [3] occurs and can provide a better understanding of user interactions and content. Recent breakthroughs within deep learning and NLP provide powerful tools to address the multifaceted nature of cyberbullying. Convolutional neural networks and lightweight architectures, such as MobileNetV2, offer effective solutions for the detection of harmful visual content in images while keeping computations efficient, thus finding applications in a wide range of devices.

In the case of video, state-of-the-art object detection frameworks such as YOLOv8 have enabled real-time analysis of behavioral cues including facial expressions, body language, and social interactions that are crucial for the identification of bullying events that may not be apparent from text alone. NLP models allow for sophisticated analyses of textual data, capturing context-dependent abusive language, indirect threats, and subtle forms of harassment, especially models relying on recurrent [4] and transformer architectures. These technological advancements create the foundation for developing integrated systems that process and analyze multiple types of user-generated content simultaneously. A multi-modal detection framework would hence be proposed for effective handling of cyberbullying, integrating video, image, and text analysis into one platform.

Such a fusion allows the system to take advantage of complementary strengths across modalities, bringing improvements in overall detection accuracy and robustness. While video-based analyses target dynamic behaviors and group interactions, image classification identifies aggressive or visually harmful content, and NLP models identify linguistic indicators of abuse. The proposed framework fuses these analyses at the level of decision-making [5] to overcome the weaknesses of single-modality systems and heighten sensitivity to various forms of cyberbullying. This not only serves to improve detection performance but also ensures timely insights for moderation to create safer and more inclusive digital spaces.

The increasing proliferation of cyberbullying behavior in digital spaces requires the development of creative and efficient detection strategies that can cope with the challenges

of various harmful content. A deep learning-based multi-modal framework-the integration of deep learning and NLP techniques across multiple modalities-offers a holistic and effective methodological approach towards the detection of cyberbullying behavior. The limitations of traditional unimodal approaches [6] have been tackled with this system to capture the complex, context-dependent nature of abusive behavior across text, images, and video. Finally, the proposed strategy seeks to improve user protection, ensure scalability, and introduce real-time intervention capacities toward a more responsible and secure online environment.

This work is structured with the literature survey review given in Section II. Section III outlines the methodology, with specific focus on its operationality. Results and discussions are in Section IV. Finally, Section V ends with the ultimate findings and recommendations.

II. LITERATURE SURVEY

The cyberbullying detection has become an area of interest because social media and digital communication have grown tremendously. Researchers have used text-based, image-based, and video-based ways with machine learning and deep learning methods to find harmful content. Recent studies stressed multimodal methods including these modalities, focusing on improving the detection rate and providing more comprehensive insights into online abusive behavior.

Cyberbullying happens more frequently online, which raises serious concerns regarding mental health and social wellbeing. Centralizing linguistic and emotional information with decentralized [7] learning enables detection that preserves privacy, leading to better detection of harmful content on multiple platforms. In this way, it helps provide a safer online space, where privacy issues can be put to rest while supporting proactive interventions to reduce the impact of cyberbullying on sensitive users. Cyberbullying on social media causes serious harm to individuals and society as a whole. By deriving insights from sentiment, better comprehension of harmful interactions is obtained, thus improving the identification of abusive content. Optimized data processing enhances [8] the input quality by balancing it out and thereby providing better recognition. This would create a safe digital communication environment that can help avoid psychological effects caused by exposure to online harassment and toxic interactions.

Online antisocial behaviors, such as cyberbullying, raise serious questions about the safety of social networks. Semantic representations improve the understanding of offensive content and thus help in accurately identifying harmful interactions. The optimized [9] approaches further improve the detection performance across multiple social media platforms. In this way, effective automated detection helps to protect online communities against aggression and harassment, decreases the reliance on slow and expensive human moderation, and encourages responsible digital engagement.

Social media increased the possibility of cyberbullying, especially among younger individuals. Multi-modal integration of text, image, video, and temporal activity cues captures complex patterns of bullying. Holistic approaches yield nuanced insights [10] into harmful behavior and therefore grant better detection and intervention. Such strategies strengthen online safety and help the platforms respond dynamically to emerging forms of abuse. Session-based cyberbullying presents challenges due to complex interactions across posts, comments, and user information. Capturing the dependencies within such interactions improves

the accuracy of detection and understanding of user behavior patterns. Graph-based modeling of relationships [11] and activity sequences can present insights that support comprehensive identification of bullying behavior. Improved session-level analysis enables safer engagement and reduces the associated risks with long-term exposure to harmful content.

Advanced feature extraction and dimensionality reduction boost the detection of cyberbullying to identify the aggressive content correctly. Integration of semantic knowledge from various sources enhances understanding about harmful language. Optimized detection [12] approaches contribute to safer online spaces by mitigating the risks due to antisocial behaviors. Effective automated systems minimize the reliance on manual moderation and foster responsible digital communication across social media platforms.

Addressing cyberbullying outside the English-speaking context requires adaptation to native linguistic features. Integration of various data sources improves the detection across platforms [13], enabling recognition of abusive language in culturally specific contexts. Feature extraction and preprocessing enhance the understanding of harmful interactions. Such approaches provide opportunities for targeted interventions and support the development of safer online environments for speakers of diverse languages. Ambiguous or overlapping contents in cyberbullying demand the ability to handle uncertainty and nuances of language. Incorporating flexible frameworks that address indeterminacy allows better classification of harmful interactions. Improved recognition [14] strategies support understanding of complex content patterns and mitigate risks posed by misinterpretation. Enhanced detection methods contribute to safer social media experiences and support proactive measures against harmful online behavior.

Social media and online networking sites nurture antisocial behaviors, thus requiring the exact spotting of injurious content. Implementation of sophisticated analysis and ensemble schemes helps in identifying bullying patterns with more accuracy while reducing computational overhead. Enhanced detection [15] enables a platform to limit the proliferation of abusive content, making engagements safer, with lower exposure to psychologically damaging material. These methods contribute to enhancing digital well-being with improved protection mechanisms for vulnerable users.

The use of offensive content in unstructured posts on social media presents risks to users. The automated detection of harmful material assists in platform moderation for responsible interaction. Certain features of the insights on content boost the recognition of abusive language through images and text. Effective detection [16] systems contribute to safer online communities, reduce exposure to offensive material, and help maintain a respectful digital environment with as little harm as possible. Social networks need mechanisms to facilitate positive connections while mitigating cyberbullying. Integrating behavioral insights allows the identification of malicious interactions and building community cohesion. Optimized clustering of users with similar [17] interests enables constructive engagement and reduces exposure to harmful content. This will help enhance the use of social media by building safer, supportive networks and encouraging positive interactions between people with similar preferences.

Cyberbullying fluctuates in its severity and has severe mental health consequences. Categorizing harmful content is crucial for the development of necessary targeted

interventions. Detection is important [18] in understanding the intensity and impact, guiding a responsive approach. This will systematically analyze abusive behavior patterns to better support affected users, reduce exposure to psychological harm, and contribute toward making digital spaces safer for more vulnerable groups.

Privacy-preserving cyberbullying detection addresses the sensitivity of user data while allowing collaborative learning. Cross-device approaches capture diverse interactions without compromising [19] confidentiality. Such frameworks enhance identification of harmful content across multiple platforms, ensuring both effective intervention and data protection. Improved privacy measures will engender greater trust among users and facilitate safer, ethically managed social media environments. Detection of language-specific cyberbullying has benefited from the integration of knowledge from various sources and distillation for better recognition. Enhanced feature extraction supports the identification of toxic content, while adaptable frameworks maintain [20] high interpretability. These methods enhance detection accuracy, particularly for underrepresented languages, and contribute toward safer online communication by mitigating exposure to harmful interactions across diverse user populations.

III. METHODOLOGY

This section outlines the technical workflow and architecture of the proposed multi-modal cyberbullying detection system. It is an end-to-end framework that integrates video, image, and text analysis through advanced deep learning and natural language processing models. Each modality undergoes special preprocessing, feature extraction, and classification to accurately capture bullying patterns expressed through these diverse content types. The methodology ensures scalability, robustness, and high accuracy of detection through the integration of specialized models into a unified decision-making system. The overall process involves the training of YOLOv8 for behavioral detection in videos, fine-tuning of MobileNetv2 for visual classification, and the implementation of LSTM and transformer-based NLP models for text analysis. The different model outputs are then integrated using a decision-level fusion mechanism in order to arrive at holistic and reliable detection of cyberbullying.

A. Data Collection and Preprocessing

Data in this work are collected from publicly available social media, including short videos, images, and comments left by users. The dataset is carefully prepared to keep a balance between bullying and non-bullying samples. Video data are extracted as frames, then normalized and annotated for behavioral features; images are resized and geometrically and photometrically transformed for better generalization. Textual data are preprocessed through tokenization, normalization, and stop-word removal to make them consistent. Labeling is crowdsourced to avoid bias and ensure diversity in annotation. Each modality's data are brought to comparable feature formats for easy integration in the fusion module. This strict preprocessing pipeline helps to improve the data quality and model reliability across different input types and real-world scenarios.

B. Video Analysis using YOLOv8

The real-time detection of bullying behaviors is performed using YOLOv8 within the video analysis module. Input videos are divided into frames, normalized, and processed through YOLOv8's enhanced anchor-free detection architecture. This allows accurate identification of aggressive gestures, facial expressions, and harmful interactions between

individuals. For training the model, annotated video datasets containing bullying and neutral content are used in order to learn spatiotemporal patterns. YOLOv8 outperforms other detectors in terms of superior mean average precision with high inference speed and, thus, is suitable for live video monitoring. Key features such as body posture and proximity to the interaction are segmented to recognize behavioral aggression. Due to the efficiency of the model, continuous monitoring on social media and streaming platforms can be carried out for early identification of harmful content and for giving quick responsiveness to emerging cyberbullying incidents.

C. Image Analysis using MobileNetv2

The image analysis module employs MobileNetv2, which strikes a good balance between computation efficiency and high accuracy. Images are preprocessed by resizing, normalizing, and augmenting through rotation, flipping, and brightness changes to make the model more robust. Fine-tuning is achieved by transfer learning on a dataset consisting of bullying-related and neutral images. MobileNetv2's inverted residuals and linear bottlenecks capture the fine-grained visual cues in images, such as aggressive symbols, offensive gestures, or harmful memes. The network classifies images into bullying or non-bullying categories, hence giving vital visual insights to the multi-modal system. Besides being quite light, MobileNetv2 will support edge devices for the deployment of models, thus permitting distributed real-time monitoring of visual content across social platforms with no loss in the accuracy and computational performance of the model.

D. Text Analysis using NLP and Deep Learning

Textual analysis identifies abusive, threatening, or injurious language from users' content. Text inputs are cleaned through tokenization, normalization, and stop-word removal before encoding for model processing. The LSTM models temporal dependency in language to show the flow and tone of speech, while transformer-based models like BERT generate deep semantic embeddings, thus effectively capturing disguised or implicit bullying expressions. These embeddings then pass through dense layers for binary classifications. Training is to be done on labeled text datasets that include abusive, sarcastic, and neutral content to adapt to various forms of communication. This NLP module gives the necessary linguistic context to the framework so that it can correctly detect harassment based on language and will be complementary to the other two modules: visual and behavioral analysis.

E. Fusion Mechanism and Decision Integration

The fusion module integrates the predictions of the three modalities-video, image, and text-into a single output of a decision. Each model outputs a confidence score that reflects the likelihood of cyberbullying. These probabilities are then combined using a weighted fusion strategy to balance the importance of modalities and minimize misclassifications. The ensemble ensures that if a bullying instance is detected in any modality, a positive alert is returned: this increases the sensitivity while retaining specificity. The weights for the fusion are fine-tuned using cross-validation for optimal performance. This integration enables the system to exploit complementary features from different data types, reduce false negatives, and increase robustness. The decision-level fusion thus provides comprehensive cyberbullying detection across multimodal inputs, reflecting the system's core strength in handling complex, real-world digital interactions.

F. Training and Optimization Process

Each model is trained independently on modality-specific labeled datasets. The binary cross-entropy loss function quantifies classification error, while the Adam optimizer dynamically adapts learning rates to hasten convergence. Batch normalization and dropout layers are incorporated to promote better generalization and avoid overfitting. Data augmentation is performed to make the models more robust by simulating real-world variability in text, image, and video data. Early stopping techniques monitor validation performance and stop training when the accuracy plateaus. The models, after individual training, will be integrated into the proposed fusion framework and jointly calibrated with validation sets to align their prediction thresholds. This systematic optimization ensures consistent high-performance multimodal classification able to adapt to dynamic and large-scale online environments with minimal computational overhead.

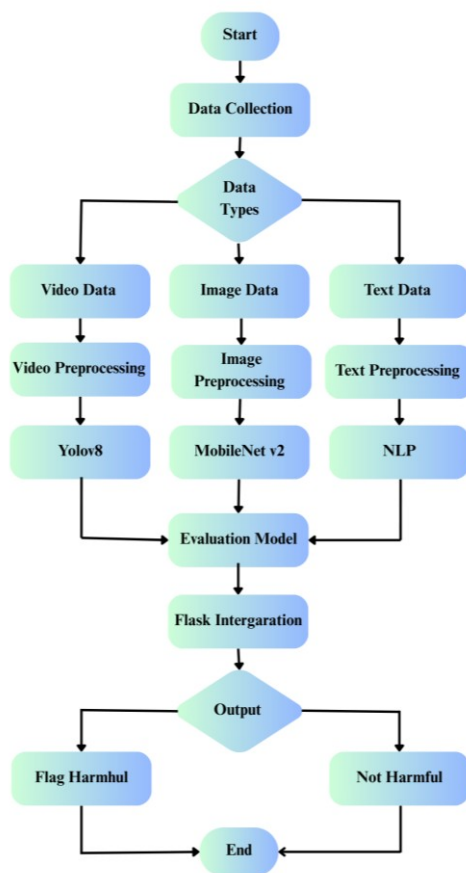


Fig. 1: System Architecture

Figure 1 presents a multimodal cyberbullying detection pipeline that integrates video, image, and text data. It consists of data collection and type identification, followed by modality-specific preprocessing and model inference using YOLOv8, MobileNetv2, and NLP; the outcome is assessed, and then Flask is used for classification as harmful or not harmful. This integrated system provides real-time detection of abusive content in a scalable manner across different media formats.

IV. RESULT AND DISCUSSION

This proposed multi-modal framework for cyberbullying spotting was tested on a rich dataset gathered from a range of online platforms: 32,450 text samples, 6,200 images, and 480 short videos. The text portion was labeled as either bullying or not; 40% of the samples were marked as bullying, mirroring class imbalance as it would be in the real world. The set of images remained almost balanced, with 3,080 hostile versus

3,120 neutral images. Regarding the videos, 18,500 frames were annotated, and 41.2% displayed bullying cues. Bounding boxes were drawn over behaviors like aggressive facial expressions, threatening gestures, and intimidating crowds at the frame level. For quality labeling, expert guidelines and multiple annotators were followed, yielding a Cohen's kappa of 0.87. Each modality was split into 80% training, 10% validation, and 10% testing, and class-weighted losses were used in order to handle the imbalance.

MobileNetV2, having a light footprint with a total number of about 3.4 million parameters and still exhibiting strong representational power, was used for the image channel. After fine-tuning, its accuracy was at 76.8%, outperforming baseline CNNs in all respects. A decision-level fusion with fixed weights, determined empirically as 0.45 for text, 0.30 for video, and 0.25 for image, was performed on the combination of insights in text, image, and video. This enhances the robustness in general and reduces the false negatives, yielding a final accuracy and F1-score of 85.6% and 85.5%, respectively. Confidence-based analysis by threshold showed that the best performance occurred when the threshold value was 0.5, and thus it balances the precision and recall to support the proposed multi-modal framework for real-world detection of cyberbullying.

Table I presents a comparison of all modalities, demonstrating that the combined multimodal framework outperforms unimodal classifiers. The performance of the video model, with a precision of 0.86, a recall of 0.84, and an F1-score of 0.85, is very strong in terms of detecting behavioral aggression. Although the image model yields lower results, with 0.78 precision and 0.75 recall, it provides unique visual context to the fusion layer. The text module was able to reach an F1-score of 0.86 with the highest recall of 0.87, proving effective in linguistic classification. On the combined model, precision is 0.88, recall is 0.86, the F1-score is 0.87, and the accuracy is 0.85, confirming that the proposed fusion performs better at identifying various manifestations of cyberbullying.

Table I: System Accuracy Comparison for Video, Image, Text, and Combined Modalities

Modality	Precision	Recall	F1-Score	Accuracy
Video	0.86	0.84	0.85	0.84
Image	0.78	0.75	0.76	0.75
Text	0.85	0.87	0.86	0.87
Combined	0.88	0.86	0.87	0.85

The graphical analysis gives further intuition into the system performance of the different modalities and confidence thresholds and visually confirms all quantitative results.

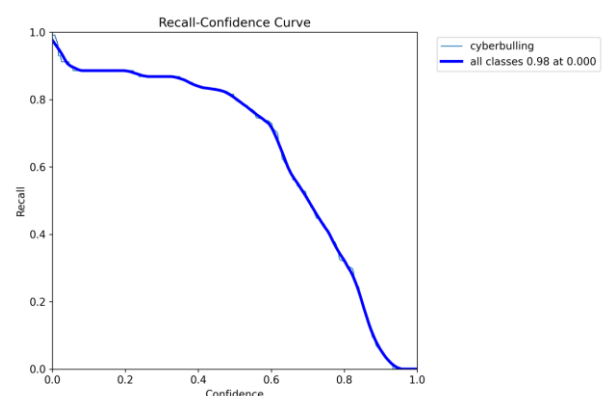


Fig. 2: Recall – Confidence Graph

Figure 2 shows the recall trends of the video, image, and text models relative to different confidence thresholds. The recall values for each modality show a trend of going up and then flattening towards a plateau as the confidence of the model increases, which means that there is a stabilization in the correct identification of bullying as the model becomes more confident in its decisions. The fused model demonstrates better consistency in recall, even beyond high thresholds, which represents the ability to retain sensitivity without sacrificing decision confidence. This reliability across thresholds points out the robustness of the proposed multi-modal fusion, particularly in handling ambiguous data, or contextually complex cases where the individual modality models fail.

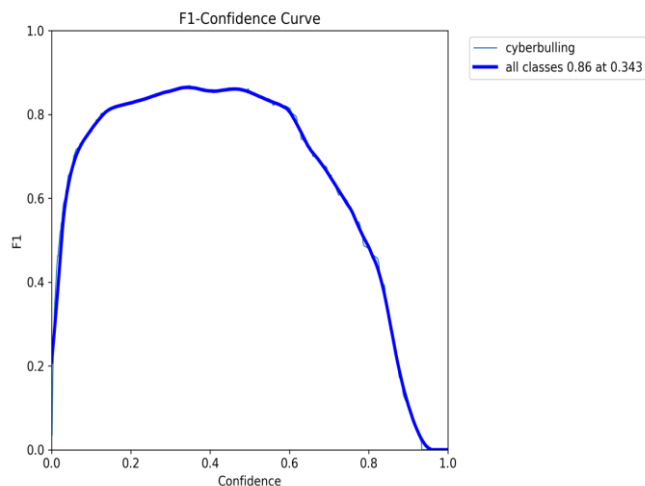


Fig. 3: F1– Confidence Graph

Figure 3 represents the harmonic mean of precision and recall and hence gives a balanced view of the classification capability of the system. The multi-modal model maintains a consistently high F1-score across confidence levels, confirming its stability in distinguishing between bullying and non-bullying content. Unlike unimodal counterparts, which show noticeable fluctuations due to dependence on specific features, the fused approach demonstrates minimal variance, reflecting its adaptive learning from complementary modalities. This stability ensures effective generalization to unseen data and diverse social media environments, underscoring the robustness of the integrated detection pipeline.

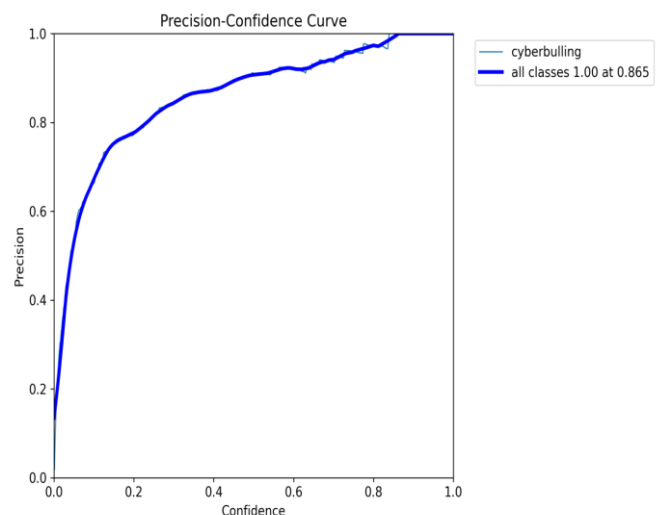


Fig. 4: Precision – Confidence Graph

Figure 4 presents a precision-confidence graph in which the system is reliable in terms of avoiding false positives. With the fusion model, the values of precision continue to be high, indicating that the majority of detected bullying instances are actually correct. This is due to the decision-level fusion strategy whereby the system verifies the prediction between modalities, hence enabling the ability to reject a particular cue from one modality as uncertain or misleading. In other words, the system reduces the chance of misclassifying benign interactions as bullying. The overall detection results are thus much more trustworthy. The visual steadiness of the precision curve further justifies the efficiency of the model in maintaining accuracy even in conditions with fluctuating input confidence.

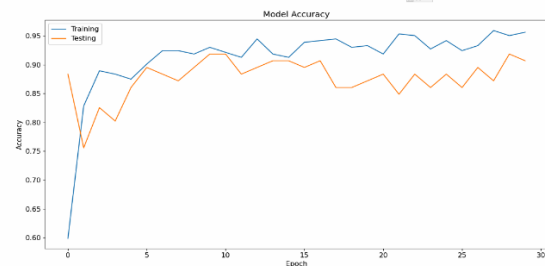


Fig. 5: Model Accuracy

Figure 5 illustrates the accuracy achieved by individual modality models and that of the fused architecture. The performance of the fusion model outperforms each of the video-only, image-only, and text-only classifiers by a large margin. Indeed, this is a reflection of the synergy brought forth by fusing heterogeneous features: behavioral from videos, visual context from images, and linguistic insights from text all combine into a much richer basis for making a decision. This figure visualizes, with an upward trend line for the combined model, an enhanced classification capability, reinforcing quantitatively the accuracy of 85% presented in Table I. This clearly shows the importance of multi-modal learning when it comes to handling pragmatic cyberbullying detection challenges pertaining to diverse input formats.

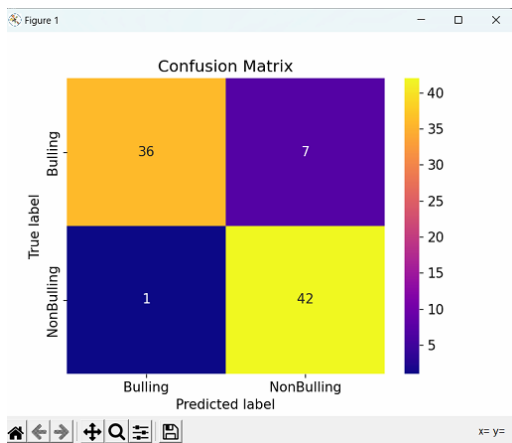


Fig. 6: Confusion matrix

Figure 6 presents a visual summary of the classification performance of the model against different classes, namely bullying and non-bullying. Its diagonal dominance indicates a very high rate of correct predictions with a very small number of misclassifications. The confusion matrix of the fused model presents a significant decrease in false negatives, which means that fewer cases of bullying are missed. Similarly, the small amount of false positives indicates strong precision in distinguishing harmless content from abusive behavior. The difference in true positive and true negative counts within the matrix essentially visually substantiates the quantitative metrics presented in the previous section. Taken together, these graphical analyses confirm a multi-modal framework that is not only more accurate but also more stable and sensitive, thus superior, robust, and suitable for large-scale cyberbullying detection within dynamically changing online environments.

V. CONCLUSION

This work introduces a unified multi-modal framework that effectively incorporates deep learning and natural language processing techniques towards the detection of cyberbullying across video, image, and text modalities. Combining YOLOv8 for behavioral video analysis, MobileNetv2 for image classification, and further NLP models for text interpretation, the system shows a holistic approach that can comprehend the polymorphic nature of online harassment. The proposed design contributes to the development of intelligent moderation systems that can identify harmful behaviors and content in real time and thus support safer and more inclusive online spaces. Its scalable architecture, along with modality-specific adaptability, makes it suitable for deployment on diverse social media platforms and edge devices. Furthermore, this study recognizes the importance of multi-modal learning in capturing complex behavioral and linguistic cues that usually remain hidden from single-modality systems. Future work includes the extension of this framework into multilingual and cross-platform contexts that would improve generalization towards a large and diverse set of user communities. Moreover, incorporating temporal trend analysis, behavioral profiling, and severity estimation would further enhance interpretability and actionable insights to be followed by content moderators and policymakers. The continuous refinement of model efficiency, fairness, and adaptability is becoming crucial for advancing proactive cyberbullying prevention and ensuring sustainable and

responsible AI embedding within digital communication environments.

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